

```
import pygame
```

```
class UI:
```

```
    def __init__(self):
```

```
        self.font = pygame.font.SysFont(
            "arial",
            18
        )
```

```
        self.title_font = pygame.font.SysFont(
            "arial",
            26
        )
```

```
    def draw_text(self, screen, text, x, y):
```

```
        surface = self.font.render(
            text,
            True,
            (230,230,230)
```

```
)
```

```
screen.blit(  
    surface,  
    (x,y)  
)
```

```
def draw_panel(self, screen, x, y, width, height):
```

```
    pygame.draw.rect(  
        screen,  
        (35,35,45),  
        (x,y,width,height),  
        border_radius=10  
    )
```

```
def draw(self, screen, robot, world,telemetry):
```

```
# Right side panel
```

```
panel_x = screen.get_width()-300
```

```
self.draw_panel(  
    screen,  
    panel_x,  
    20,  
    280,  
    500  
)
```

```
# Title
```

```
title = self.title_font.render(  
    "ASTERIA",  
    True,  
    (0,200,255)  
)
```

```
screen.blit(  
    title,  
    (panel_x+20,35)  
)
```

```
# Robot information
```

```
y = 90
```

```
self.draw_text(  
    screen,  
    "ROBOT STATUS",  
    panel_x+20,  
    y  
)
```

```
y+=35
```

```
self.draw_text(  
    screen,  
    f"Position: {int(robot.x)}, {int(robot.y)}",  
    panel_x+20,  
    y  
)
```

```
y+=25
```

```
self.draw_text(  
    screen,  
    f"Speed: {robot.speed}",  
    panel_x+20,  
    y  
)
```

```
y+=25
```

```
self.draw_text(  
    screen,  
    f"Direction: {robot.angle} deg",  
    panel_x+20,  
    y  
)
```

```
y+=50
```

```
self.draw_text(  
    screen,  
    f"Angle: {robot.angle} deg",  
    panel_x+20,  
    y  
)
```

```
    screen,  
    "ENVIRONMENT",  
    panel_x+20,  
    y  
)
```

```
y+=35
```

```
self.draw_text(  
    screen,  
    f"Temperature: {world.temperature} C",  
    panel_x+20,  
    y  
)
```

```
y+=25
```

```
self.draw_text(  
    screen,  
    f"Luminosity: {world.luminosity}",  
    panel_x+20,  
    y
```

```
)
```

```
y+=25
```

```
self.draw_text(  
    screen,  
    "Terrain: Unknown",  
    panel_x+20,  
    y  
)
```

```
y+=50  
self.draw_text(  
    screen,  
    "SENSOR DATA",  
    panel_x+20,  
    y  
)
```

```
y += 35  
self.draw_text(  
    screen,  
    f"Front:{robot.front_distance}cm",  
    panel_x+20,
```

```
    y  
)
```

```
y += 25
```

```
self.draw_text(  
    screen,  
    f"Rear:{robot.rear_distance}cm",  
    panel_x+20,  
    y  
)
```

```
y += 25
```

```
self.draw_text(  
    screen,  
    f"Left:{robot.left_distance}cm",  
    panel_x+20,  
    y  
)
```

```
y+=25
```

```
self.draw_text(  
    screen,  
    f"Right: {robot.right_distance}cm",  
    panel_x+20,  
    y  
)
```

```
y +=25
```



```
self.draw_text(screen,f"Zone:{telemetry['Zone']}",panel_x+20,y)
```

```
y+=25
```

```
self.draw_text(screen,f"Risk:{telemetry['Status']}",panel_x+20,y)
```